

LAB 3 – WRITTEN JUSTIFICATION

COMMUNITY SOLAR CREDIT ALLOCATION MANAGER

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1. Architecture Selection

We chose Layered Architecture for the Community Solar Credit Allocation Manager. The system is divided into Presentation Layer, Business/Application Layer and Data Layer. This structure is suitable because the system has different activities such as user interaction, solar generation processing, credit transfer, billing offset calculation and data storage.

2. Architectural Choice

The Presentation Layer contains the Prosumer Interface and Co-op Manager Interface. It is responsible for receiving requests and displaying information to the users. The Business/Application Layer contains Authentication & Authorization, Solar Generation Processor, Credit Transfer Manager and Billing Offset Manager. These components perform the main system operations. The Data Layer contains the Solar & Credit Database, which stores solar generation data, credit balances, transactions, billing offsets and user information. This gives a clear separation between the user interface, processing logic and stored data.

3. Reason 1 – Separation of Responsibilities

The system has several different responsibilities and they should not be handled by a single component. The Solar Generation Processor receives and validates hourly kWh data from the Smart Meter System. The Credit Transfer Manager handles transfer of surplus credits between Prosumer Resident accounts. The Billing Offset Manager calculates and tracks monthly utility billing offsets. Keeping these functions separate makes the system easier to understand, maintain and modify. For example, changes in billing calculations can be made in the Billing Offset Manager without changing the solar generation processing component.

4. Reason 2 – Controlled Credit Transfer

Credit transfer is one of the main operations of the system. Before a transfer is processed, the system must check the user authentication, validate the neighbor account and verify that the Prosumer Resident has enough available credits. After confirmation, the transferred credits must be deducted from the sender and added to the neighbor, and the transaction must be recorded. The Credit Transfer Manager provides one controlled place for these operations. This helps to keep both account balances correct and prevents invalid transfers.

5. Security Advantage

The architecture provides a separate Authentication & Authorization component. A Prosumer Resident must be authenticated before initiating a solar credit transfer. The Credit Transfer Manager can perform an authorization check before processing the request. This directly supports NFR-002, which requires that only authenticated Prosumer Residents can initiate transfers. Separating authentication from the user interface also reduces direct access to sensitive credit information and makes access control easier to manage.

6. Performance Benefit

The Solar Generation Processor can process smart-meter telemetry separately from normal user interface operations. Hourly generation data can be validated and stored without making the Prosumer Interface responsible for the complete processing task. This separation helps the system handle generation data from the required 500 households and supports the stated 99.9% uptime requirement. The database can also be organized for efficient access to generation records, balances, transactions and billing offsets.

7. Component and Interface Summary

Component	Main Responsibility	Interface / Interaction
Prosumer Interface	View balance and transfer credits	Authentication API / Credit Transfer API
Co-op Manager Interface	View system and billing information	Billing Offset API
Authentication & Authorization	Authenticate and authorize users	Authentication Service
Solar Generation Processor	Validate and process hourly kWh	Telemetry / Generation Data
Credit Transfer Manager	Validate transfers and update balances	Credit Transfer Service
Billing Offset Manager	Calculate and track monthly offsets	Billing Offset Service
Solar & Credit Database	Store generation, balances and transactions	Database Interface

8. Conclusion

Layered Architecture provides a clear and organized design for the Community Solar Credit Allocation Manager. It separates presentation, business processing and data storage, while allowing authentication, credit transfer, solar telemetry and billing operations to be handled by dedicated components. Therefore, the selected architecture addresses the main functional, security and performance requirements of the given problem.